AI-Based Project Proposal
Project Title: **UrbanSightAI**

Course Code : SE 803

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ProjectTitle:**UrbanSightAI**

Data-Driven Governance for Resilient Cities

1. Introduction & Problem Statement

Rapid urbanization in Bangladesh has exposed governance shortfalls that are the principal obstacle to realizing SDG-11 at the local level. Core municipal functions,drainage,road maintenance, land-use regulation, utilities and emergency response are split among multiple agencies (city corporations,RAJUK,WASA,LGED), producing overlapping mandates, duplicated investments and slow inter-agency handoffs that stall implementation at the ward level [1][4]. At the same time, municipal fiscal autonomy is weak and planning capacity remains limited; critical ward-scale operational data are fragmented, outdated, or inaccessible to the staff who must make budget decisions [2][5]. Informal settlements where vulnerability concentrates are frequently unmapped or excluded from routine plans, leaving large populations uncounted in resource allocation and exposed to repeated service failures (waterlogging, fire risk, sanitation gaps) [14][15].

This is a governance problem more than a technical one: councils and engineers rarely lack willing action; they lack trusted, ward-level evidence, standardized decision products, and straightforward budget-ready justification that fit municipal procurement and financing cycles. Without these policy instruments, authorities default to reactive, costly interventions rather than preventive, prioritized investments that reduce exposure and deliver durable SDG-11 outcomes [1][20]. UrbanSightAI is explicitly designed to close that gap by converting spatial data and field verification into budget-aware, interagency-ready policy outputs (Top-N worklists, costed memos, SDG-11 ward cards) municipal teams can adopt immediately [17][18][19].

2. Motivation & Rationale

Policy impact requires alignment of three elements: evidence, decision rules, and financing. In Bangladesh, two persistent bottlenecks impede that alignment: (a) inconsistent or inaccessible ward-level evidence and (b) constrained, opaque municipal finance that slows targeted action [2][5]. UrbanSightAI is designed to bridge these gaps with a “policy-first” architecture: AI and geospatial analytics are used not as ends but as translators converting imagery and reports into plain-language, costed, implementable actions that match municipal workflows and budget cycles [1][13].

Why this matters for policy. Simple, low-cost micro-interventions (drain cleaning, short road repairs, waste hotspot clearance, targeted tree planting) deliver outsized returns when allocated to the right places. But planners lack a repeatable, auditable method to pick the “right places.” UrbanSightAI provides that method: it produces ranked, equity-weighted intervention lists, shows the expected beneficiaries, flags permits and responsible agencies, and packages all this

into procurement-ready briefs for finance and engineering teams. Evidence from municipal observatories and participatory platforms shows that decision-support tools which produce exportable, accountable policy products are the ones that get used in budgets and council meetings; the design therefore prioritizes usability and defensibility over maximum technical sophistication [11][13][8].

Operationally, the pilot uses light, accessible data (Sentinel/Landsat, municipal shapefiles or OSM fallbacks, and focused field labeling of ~200–400 geotagged points) so municipalities can implement quickly without heavy sensor investments [17][18][19]. Civic validation (Bangla chatbot + participatory portal) is embedded so community priorities appear as weighted inputs in ranking increasing legitimacy and reducing political friction [8][10]. Economically, prioritizing micro-works reduces repeated emergency spending and creates clear, data-based cases for reallocation or donor support, a pragmatic path to scale that speaks directly to finance officers and councillors [5][6]. The pilot will demonstrate measurable policy wins (reduced detection-to-workorder time; share of Top-N items adopted), creating a replicable model for SDG-11 localization across Bangladeshi cities [1][20].

3. Background Study / Related Works (policy-oriented synthesis)

UrbanSightAI draws on three strands with direct policy relevance.

SDG localization & municipal finance:UN inter-agency policy briefs and national reviews emphasize that SDG delivery is local by design: achieving SDG targets requires empowered municipal governments with interoperable data, fiscal tools, and transparent prioritization mechanisms [1][2][20]. Studies of subnational finance document limited own-source revenue and dependency on central transfers, which delays action even when priorities are known, underscoring the need for tools that improve the case for timely municipal spending [5][4].

Operational urban platforms & participatory governance: Global practitioner cases show how fused data plus governance-focused outputs change municipal behavior. Jakarta's flood dashboard and city observatories illustrate how data integration informs operational decision-making; Decidim and civic chatbots demonstrate how structured citizen input can be turned into traceable policy signals used in budgeting and planning [11][13][8][9][10]. The core lesson: dashboards that remain purely informational rarely change budgets but systems that produce simulation outputs, procurement-ready briefs and participatory validation do affect policy.

Technical evidence for implementable indicators: Remote-sensing and ML research confirm that feasible, operational indicators (informal settlement footprint, blocked-drain counts, green-deficit indices) can be generated from multispectral imagery when paired with targeted field validation and simple spatial joins [14][15]. Short-horizon predictors and change-detection

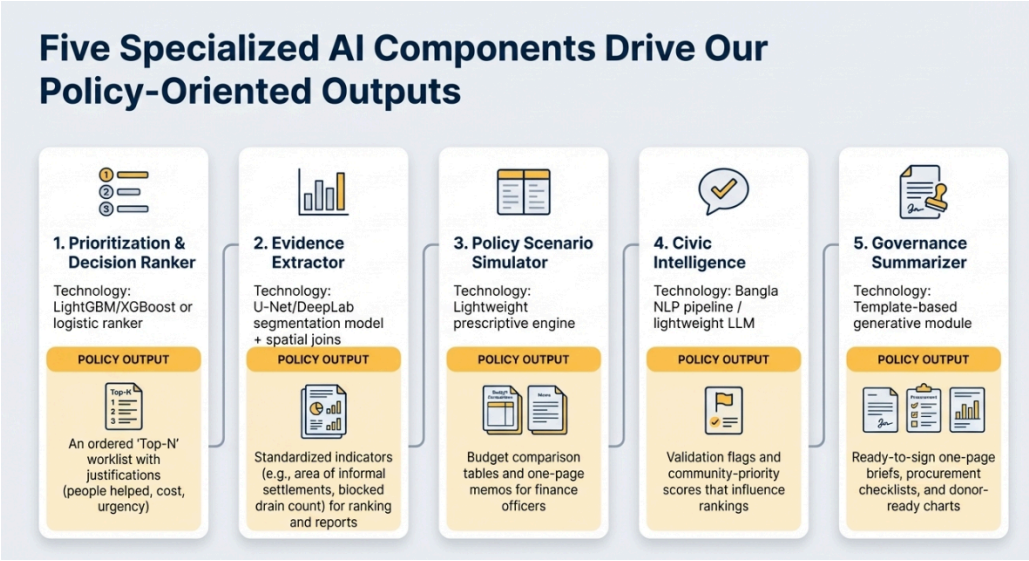
methods add timing and evidence for urgency, but their policy value depends on packaging into budgeted, interagency actions rather than raw forecasts [16]. Few projects close this last mile: convert mapping into cost, equity-aware intervention lists and interagency coordination packets which is precisely the gap UrbanSightAI addresses by design [12][14][15].

4.0 Proposed Solution Technical Methodology.

UrbanSightAI is a policy-first platform designed to:

i. Equip municipal decision-makers with a ward-level digital twin that summarizes assets, service gaps, exposure and governance constraints.
ii. Produce ranked, cost policy actions (micro-works) formatted for budget requests, procurement and engineering work orders.
iii. Run scenario simulations that show budget → outcome tradeoffs (impact, equity, timeline).
iv. Integrate citizen evidence (chatbot / participatory portal) so community priorities and complaints directly influence policy ranks.
v. Provide automated SDG-11 governance cards, inter-agency packets and audit trails for transparency and donor reporting.

AI Components policy tools (what they do, and what they output)



1. Prioritization & Decision Ranker (core)

   What it is: A ranker (LightGBM / XGBoost or logistic ranker) that scores candidate projects by impact-per-cost, feasibility and equity weighting.

   Policy output: Ordered “Top-N” worklist with simple justifications (people helped, cost, why now).

2. Policy Scenario Simulator (prescriptive engine)

   What it is: Lightweight simulator that tests counterfactuals (allocate X BDT → predicted reduction in affected households).

   Policy output: Budget comparison tables and one-page memos for finance officers.

3. Evidence Extractor (segmentation → indicators)

   What it is: Segmentation model (U-Net/DeepLab) + spatial joins that convert imagery and vectors into policy indicators (area informal, blocked drains count, green deficit).

   Policy output: Standardized indicators used in ranking and reports.

4. Civic Intelligence (NLP + Chatbot)

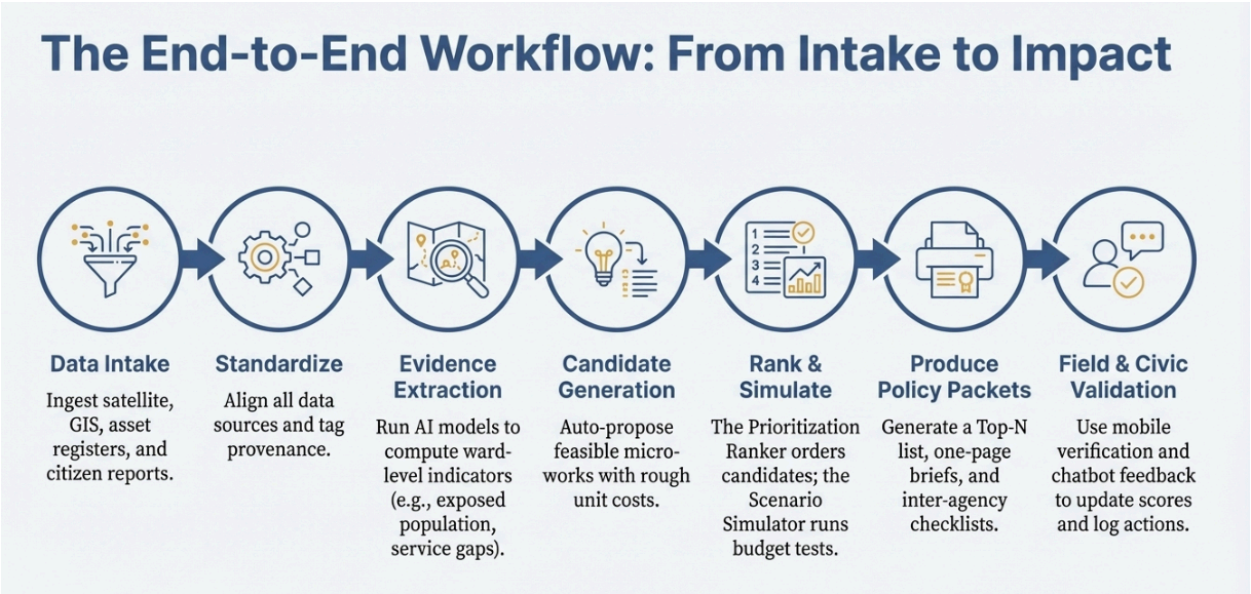
   What it is: Bangla NLP pipeline / lightweight LLM that classifies citizen reports, sentiment and priority signals.

   Policy output: Validation flags and community-priority scores that nudge rankings and create community-facing summary pages.

5. Governance Summarizer (generative brief generator)

What it is: Template-based generative module that converts data + ranking into editable policy briefs and procurement notes.

Policy output: Ready-to-sign one-page briefs, procurement checklists and donor-ready charts.



Technical Workflow concise, policy-oriented steps

1. Data intake: ingest satellite, ward GIS, asset registers, census/gridded pop, past workorders, citizen reports.

2. Preprocess & standardize: tile imagery, compute indices, align vectors and tag provenance.

3. Evidence extraction: run segmentation and anomaly detectors → compute ward indicators (exposed population, service gaps).

4. Candidate generation: auto-propose feasible micro-works (jurisdiction, permit flag, rough unit cost).

5. Rank & simulate: Prioritization Ranker orders candidates; Scenario Simulator runs budget tests and equity impacts.

6. Produce policy packets: Top-N list → one-page brief + interagency checklist + export (CSV/PDF) for procurement.

7. Field & civic validation: mobile verification + chatbot feedback update scores; system logs actions for audit and retraining.

Full-Scale Future Implementation (beyond course pilot)

i. Deploy microservices for resilient, multi-ward scaling.
ii. Integrate with municipal financial systems and procurement APIs for end-to-end budget workflows.
iii. Expand indicator library (health, mobility, air/heat metrics) and multi-season training data.
iv. Federate municipal datasets with national LGD/RAJUK resources under MOUs.
v. Add role-based governance console (finance, engineering, councillor views) and meeting-ready export packs.
vi. Institutionalize a small UrbanSightAI policy unit inside city government to steward data, priorities and model updates.
vii. Offer managed service model (municipal subscription + capacity building) to fund continued operations.

5.0 Scope and Pilot Implementation

Project Scope

i. Build the core UrbanSightAI system with an AI Prioritization engine, Digital-Twin assembly, Bangla chatbot (citizen input), and a Policy Scenario Simulator.
ii. Provide chatbot interfaces and citizen-facing support (chatbot messages, participatory portal summaries, and simple field forms).
iii. Deliver a web planner dashboard for ward selectors, SDG-11 cards, exportable Top-N worklists and policy packets.
iv. Train and validate models using small-scale, locally collected datasets in the pilot (satellite patches, 200–400 geotagged field photos, basic municipal layers).

Pilot Implementation (For This Course)

For the course pilot we will implement a ward-scale, policy-focused proof-of-concept with these concrete components:

1. **Basic web dashboard & user registration** role-based login (planner / enumerator / viewer), ward selector, last-update stamp, CSV/PDF export.

2. **Pilot Prioritization Model** a lightweight ranking model (LightGBM baseline) that scores candidate micro-works by impact-per-cost and equity; outputs a Top-10 worklist.

3. **Segmentation & Evidence Extractor (pilot)** U-Net/transfer-learning pilot that maps informal patches and generates standardized ward indicators (exposed area, blocked-drain count).

4. **Policy Scenario Simulator (mini)** simple budget→impact simulator that tests 2–3 counterfactuals and produces one-page memos for finance officers.

5. **Chatbot & Field Data Pipe** basic NLP classifier for citizen reports (blocked drain, flood, waste) plus an ODK/KoBo form; includes a notification/simulation console to demo stakeholder flows.

The remaining modules (full-scale simulation, interagency packet automation, reinforcement-learning prioritizer, national rollout features) will be planned for post-course expansion and productization.

6. Budget & Resource Plan

Table 6.1: Estimated budget (personnel costs)

Economic code	Description	Unit	Cost per unit (lakh Tk)	Estimated cost (lakh Tk)	% of total	Justification
A1	AI/ML Engineers (Model dev, ranker, simulator)	2	1.8	3.6	18.3	Build LightGBM/XGBoost ranker, scenario simulator, and ML pipelines; core technical delivery
A2	Geospatial Analyst (Imagery + indicators)	1	1.65	1.65	8.4	Sentinel/Landsat preprocessing, U-Net mapping, indicator generation, spatial joins
A3	Backend Developer (APIs + integration)	1	1.50	1.50	7.6	Planner APIs, data intake services, export endpoints (CSV/PDF)
A4	Frontend/UI Developer (Dashboard + portal)	1	1.35	1.35	6.9	Ward dashboard, digital-twin assembly, role-based UI, low-literacy flow
A5	Policy Researcher (Briefs, SDG-11 cards)	1	1.20	1.20	6.1	Governance summaries, procurement-ready memos, inter-agency packet drafting
A6	Field Enumerators (Citizen input, validation)	2	0.75	1.5	7.6	200–400 geotagged points, ground-truthing, ODK/KoBo ops and QA
				Subtotal personnel (A): 10.80	54.8	

Table 6.2: Estimated budget (operational expenses)

Economic code	Description	Unit	Cost per unit (lakh Tk)	Estimated cost (lakh Tk)	% of total	Justification
B1	Cloud hosting (VM + modest GPU)	1	0.60	0.60	3.0	Training/inference for pilot models; uptime for planner/dashboard
B2	Data acquisition (satellite patches, field photos)	1	0.75	0.75	3.8	Imagery tiles, labeled samples, municipal layers (OSM fallbacks)
B3	Software tools/licenses	1	0.40	0.40	2.0	GIS add-ons, design tools, paid plugins (major ML libs remain free)
B4	Communication & outreach (chatbot/SMS, portal)	1	0.50	0.50	2.5	SMS quotas, portal messaging, citizen feedback operations
B5	Training & workshops (municipal + enumerators)	1	0.60	0.60	3.0	Onboarding, hands-on sessions, materials for operational handoff
				Subtotal operational (B): 2.85	14.5	

Table 6.3: Estimated budget (capital expenditure)

Economic code	Description	Unit	Cost per unit (lakh Tk)	Estimated cost (lakh Tk)	% of total	Justification
C1	GPU-enabled workstation/server	1	2.00	2.00	10.1	Local ML training, reproducible experiments, secure data handling
C2	Field tablets/smartphones (enumerators)	4	0.25	1.00	5.1	Reliable field capture, offline forms, photo evidence collection
C3	Networking & storage equipment	1	0.50	0.50	2.5	NAS/backup, faster internal transfers, dataset persistence
				Subtotal capital (C): 3.50	17.8	

Table 6.4: Estimated budget (administrative, overhead and risk buffer)

Economic code	Description	Cost (lakh Tk)	% of total	Justification
D1	Institutional overhead (10% of A+B+C)	1.72	8.7	Standard overhead on direct costs (A+B+C = 17.15 lakh)
D2	Contingency / risk buffer (2% of A+B+C)	0.34	1.7	Field and ops uncertainties (weather delays, data gaps)
D3	Miscellaneous administrative costs	0.50	2.5	Procurement fees, printing, local compliance, small gear
	Total administrative (D)	2.56	13.0	

Consolidated budget

- **Personnel (A):** 10.80 lakh
- **Operational (B):** 2.85 lakh
- **Capital (C):** 3.50 lakh
- **Administrative/Overhead (D):** 2.56 lakh

TOTAL PROJECT COST (A + B + C + D): 19.71 lakh Taka

7. Impact on Society

UrbanSightAI strengthens urban governance by enabling data-grounded policy, fair budgeting, and transparent decision-making. It will:

i. Improve ward-level planning through actionable, quantified insights.
ii. Reduce resource waste by replacing reactive fixing with evidence-based prioritization.
iii. Enable equitable development by highlighting underserved zones in policy ranking.
iv. Increase transparency through citizen participation and audit-ready decision logs.
v. Support faster response, better spending, and fairer city service distribution.

This leads to more accountable institutions, safer urban environments, and resilient growth.

8. Alignment with Sustainable Development Goals (SDGs)

UrbanSightAI contributes directly to **SDG-11: Sustainable Cities & Communities**, and supports several governance and climate-aligned targets.

SDG 11: Sustainable Cities & Communities

- Target 11.3 Improve inclusive and sustainable urban planning.
- Target 11.5 Reduce disaster impact through early signals & resource planning.
- Target 11.6 Better waste/drain prioritization reduces environmental burden.

- Target 11.7 Helps planners increase access to public spaces & green zones.
- Target 11.b Supports risk-sensitive policies via data-driven planning tools.

Additional SDG Linkages

- SDG 13: Climate Action Supports climate-risk planning & impact-forecasting.
- SDG 9: Industry, Innovation & Infrastructure Builds digital governance capacity.
- SDG 16: Peace, Justice & Strong Institutions Promotes transparency, accountability & citizen-inclusive decision-making.

9.0 Revenue Plan & Sustainability

UrbanSightAI is designed to be socially useful and financially sustainable. The revenue model combines municipal subscriptions, data services, consultancy, grants and light civic premium options. Each stream keeps basic civic access free while generating stable income to fund operations, model updates and scale.

1. Municipal Subscription (Core B2G / B2City)

What: Tiered annual subscriptions by city size / wards (basic → standard → enterprise). Includes dashboard access, Top-N worklists, scenario simulator, API access and admin training.
Why it generates revenue: City budgets already fund planning tools; a small line-item for data-driven prioritization is cost-justified by improved budget efficiency and reduced emergency spend.

2. Data-as-a-Service (DaaS) & API Access

What: Pay-per-dataset or subscription API (ward indicators, historical exposure maps, time-series exports) for planners, researchers and NGOs.
Why: Researchers and consultancies pay for clean, processed urban datasets; recurring API fees scale with usage.

3. Consultancy & Integration Services

What: One-time fees for integration (procurement packet automation), training workshops, MOU facilitation, and custom model tuning.

Why: Municipalities lack in-house capacity; paid integration reduces adoption friction and funds local onboarding.

4. Grants & Development Funding

What: Seed grants and implementation funding from donors, development banks, and UN agencies for pilot and scale phases.
Why: Development partners favor measurable SDG tools; grants lower municipal up-front costs and de-risk adoption.

5. Performance-Based / Results Contracts

What: Contracts that tie fees to agreed outcomes (e.g., % of Top-N actions adopted, reduction in detection→workorder time).
Why: Demonstrates value and aligns incentives donors or city units pay more when measurable improvements occur.

6. Civic Premium Services (Optional & Low-Cost)

What: Basic citizen reporting and dashboard remain free. Optional paid civic features (neighborhood alerts via SMS, advanced historical reports for NGOs) offered at low cost.
Why: Maintains accessibility while monetizing high-value convenience features for local NGOs and community groups.

7. Ethical Commercial Partnerships

What: Aggregated, anonymized analytics for utilities, insurers, or real-estate firms (non-sensitive, opt-in). No individual-level or household-identifying commercial sales.
Why: Private sector will pay for city-level risk and exposure analytics if privacy and ethics are preserved.

8. Training & Certification Programs

What: Paid certification for municipal staff and consultants in using UrbanSightAI workflows and interpreting outputs.
Why: Helps institutionalize use and creates a recurring training revenue stream.

9. White-label / Licensing

What: License the platform (or parts) to regional consultancies or NGOs who want a branded deployment.
Why: Accelerates geographic scale without full in-house expansion.

10. Expected Outcomes and Deliverables

1. AI Policy Prioritization Output

A trained ranking model that scores micro-works by impact, cost, and feasibility.
Generates a Top-N intervention list for planners with brief justifications.

2. AI Governance Chatbot

A functioning Bangla chatbot for citizen reporting and feedback.
Provides verified service information and turns complaints into structured policy signals.

3. Ward-Level Digital Twin Dashboard

An interactive map showing informal settlements, service gaps, road/drain conditions.
Helps identify where interventions are most needed visually, instantly.

4. Budget Scenario & Policy Recommendation Engine

A prescriptive simulation tool that estimates outcome change per budget spent.
Suggests low-cost micro-projects with population benefit and timeline estimation.

5. Integrated Web Platform for Planners

Fully working dashboard where urban officials can:
1.view risk & indicators
2.compare interventions

3.export policy briefs (PDF/CSV)
Backend APIs connect all AI components into one cohesive governance tool.

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